

Lab07_tavares

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(DBI)
library(RSQLite)

# Conectando ao banco local e armazenando na variável conn
conn <- dbConnect(RSQLite::SQLite(), "uwmadison.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

```
SELECT
    COUNT(DISTINCT i.name) AS quantidade_professores,
    GROUP_CONCAT(DISTINCT i.name) AS nomes_professores
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections s ON t.section_uuid = s.uuid
JOIN course_offerings co ON s.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT';
```

quan-
ti-
dade_pro-
fes-
somes_professores

Table 1: 1 records

quan-
ti-
dade_pro-
fes-
somes_professores

377 SHENG WANG, DONALD PORTER, MINJING TAO, KUNLING HUANG, JARED BROWN, DONG XIA, JENNIFER NGUYEN, YUCHEN ZHOU, SEULKEE YUN, YONGFENG WU, VICTOR LUO, DUZHE WANG, JOHN GILLETT, DUY NGUYEN, BROOK LUERS, GUN WOONG PARK, YAOYAO XU, JIE ZHANG, QUOC TRAN, SHANE HUBLER, LONG PHAN, WESLEY CHANG, TONG LI, FAN YANG, JU HEE CHO, JEEA CHOI, SHIXUE LIU, BIN DAI, XIPEI YANG, DOUGLAS M. BATES, MICHAEL GEORGE ILLIS, WEI-YIN LOH, CRYSTAL CHEN, JILI WANG, LIAM JOHNSTON, YIFAN MEI, RUI CHEN, JIE SONG, LUWAN ZHANG, GINA BENNINGER, HAODA FU, XIWEN MA, ERICA LEE DEADMAN, JINGJIANG PENG, DEYUAN JIANG, MENG SONG, AKICHIKA OZEKI, HAO TENG, GONZALO CONTADOR, XIN ZHANG, DONGGYU KIM, FREDERICK BOEHM, YONGSU LEE, COLIN LONGHURST, BO YANG, HAO CHEN, YI LI, CLAIRE BOBST, SEAN KENT, RUNGANG HAN, DEBRAJ DAS, CHAN PARK, SHENGJI JIA, SIJING LI, YUCHANG WU, ZHUANG WU, YOUNG MIN PARK, HYEBIN SONG, BIN ZHANG, XIAOMAO LI, YOURAN QI, QIURONG CUI, SEHO PARK, SHENG ZHANG, HUIKUN ZHANG, ALYSSA DIGILIO, YUQING XU, STEPHEN BERG, PEIGEN ZHOU, YAN CHEN, QIONG ZHANG, YOUNGDEOK HWANG, HAN CHEN, YUJIN CHUNG, ZHENGXIAO WU, NICHOLAS STEPHEN KEULER, WENWEN ZHANG, PERLA REYES, ISMOR FISCHER, XINWEI DENG, YANG ZHAO, JIAJIE CHEN, CHIA-CHIEH LIN, ZUOFENG SHANG, QI TANG, MING XIE, JIALE XU, JINGCI MENG, BIN ZHU, QUEFENG LI, HAO ZHENG, XU XU, JUN LI, YU QIU LIU, ANQI SHI, KYLE HEBERT, FAN GAO, XIUYU MA, VICTORIA MANSFIELD, XINYU SONG, YUANZHI LI, XIAO NIE, CHEN CHENG, YI LIU, ANDREW LESLIE, CHANHAN HSU, YAOGUO XIE, TIMOTHY IDOWU, KAM-WAH TSUI, KEVIN PACKARD, CHENLIANG XU, JOSEPH DEUTSCH, RYAN ZEA, WENZHI CAO, LAN LUO, ROBERT WARDROP, SEUNGBONG HAN, SANGBUM CHOI, RUI TANG, ZHIGUO XIAO, XIN LI, RICHARD A. JOHNSON, JUN ZHANG, XU HE, GARY HOWARD SCHROEDER, BO HUANG, HEATHER MARIE BRAZEAU, IAN BRANSTAD RILEY, CHENXI LI, YUAN JIANG, JUNHEE HAN, SONA Z SWANSON, CHEN JING, ZIFENG ZHAO, CUIZE HAN, JINGLAN LI, YING ZHANG, LILUN DU, XIRAN WANG, JOHN DAVIS, YI CHAI, ALAN HUANG, HAOYANG FAN, THU LE, ALEXANDER COVINGTON, LILI ZHENG, REBECCA POST, CHELSEY GREEN, TZU HSIANG HUNG, ABIGAIL BENZINE, TUN LEE NG, MICHAEL KUTZLER, BOWEN HU, LU YANG, RUOSI GUO, XUN ZHAO, MIN NIU, JIAN WU, WEI ZHENG, CLAUDIA SOLIS LEMUS, XINJIE HE, JIE WEI, XUEYAO CHEN, CHENGNING ZHANG, MANJUSHA KANCHARLA, GINA OH, XIAOWU DAI, LUXI CAO, MITCHELL PAUKNER, TIEN VO, SOHEIL SADEGHI, SONG WANG, SHIZHEN WANG, LAM HO, GENG LI, JENNIFER BIRSTLER, BINGYING XIE, MARIA KAMENETSKY, KYLE HERBET, JIANCHANG HU, TRAM TA, YUAN WANG, CHIEN-WEI CHEN, KAZUHIKO SHINKI, SIJIAN WANG, BEHZAD AALIPUR, DEREK BEAN, SCOTT JOSEPH HETZEL, BEN ADAM HAALAND, VARSHA KULKARNI, FANG FANG, MIN JUNG LEE, ELOISA D

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

I- Quem é o professor mais difícil?

```
SELECT i.name AS professor,
       SUM(g.a_count*4 + g.ab_count*3.5 + g.b_count*3 + g.bc_count*2.5 + g.c_count*2 + g.d_count*1 + g.f_count*0) /
       SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_count) AS gpa_medio
FROM grade_distributions g
JOIN sections s ON g.course_offering_uuid = s.course_offering_uuid
JOIN teachings t ON s.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
JOIN subject_memberships sm ON s.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY i.name
HAVING gpa_medio IS NOT NULL
ORDER BY gpa_medio ASC
LIMIT 1;
```

Table 2: 1 records

professor	gpa_medio
JAMES D KUELBS	2.607692

II - Quem é o professor mais fácil?

```
SELECT i.name AS professor,
       SUM(g.a_count*4 + g.ab_count*3.5 + g.b_count*3 + g.bc_count*2.5 + g.c_count*2 + g.d_count*1 + g.f_count*0) /
       SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_count) AS gpa_medio
FROM grade_distributions g
JOIN sections s ON g.course_offering_uuid = s.course_offering_uuid
JOIN teachings t ON s.uuid = t.section_uuid
JOIN instructors i ON t.instructor_id = i.id
JOIN subject_memberships sm ON s.course_offering_uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY i.name
ORDER BY gpa_medio DESC
LIMIT 1;
```

Table 3: 1 records

professor	gpa_medio
YAJUAN SI	4

III - Qual é a disciplina mais difícil?

```
SELECT co.name AS disciplina,
       SUM(g.a_count*4 + g.ab_count*3.5 + g.b_count*3 + g.bc_count*2.5 + g.c_count*2 + g.d_count) /
       SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_count) AS gpa_medio
FROM grade_distributions g
JOIN course_offerings co ON g.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY co.name
HAVING gpa_medio IS NOT NULL
ORDER BY gpa_medio ASC
LIMIT 1;
```

Table 4: 1 records

disciplina	gpa_medio
Intro-Theory of Probability	2.886374

IV - Qual é a disciplina mais fácil?

```
SELECT co.name AS disciplina,
       SUM(g.a_count*4 + g.ab_count*3.5 + g.b_count*3 + g.bc_count*2.5 + g.c_count*2 + g.d_count) /
       SUM(g.a_count + g.ab_count + g.b_count + g.bc_count + g.c_count + g.d_count + g.f_count) AS gpa_medio
FROM grade_distributions g
JOIN course_offerings co ON g.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
GROUP BY co.name
ORDER BY gpa_medio DESC
LIMIT 1;
```

Table 5: 1 records

disciplina	gpa_medio
Sample Survey Theory&Method	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

5. Data de hora

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 08:34:17 -03"
```